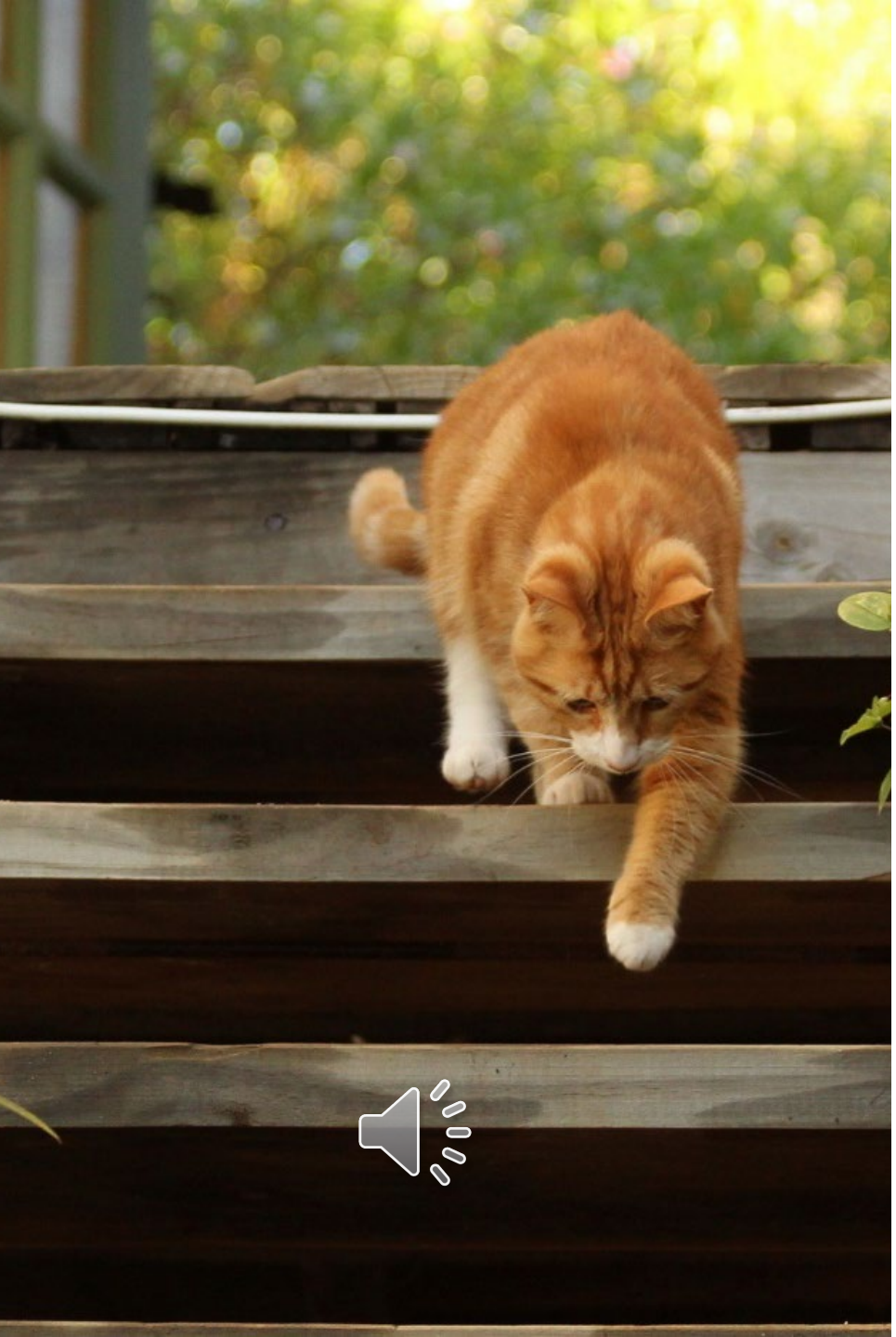
The treasure-hunter





Top-down: from audience to message

Knowing your audience will help you

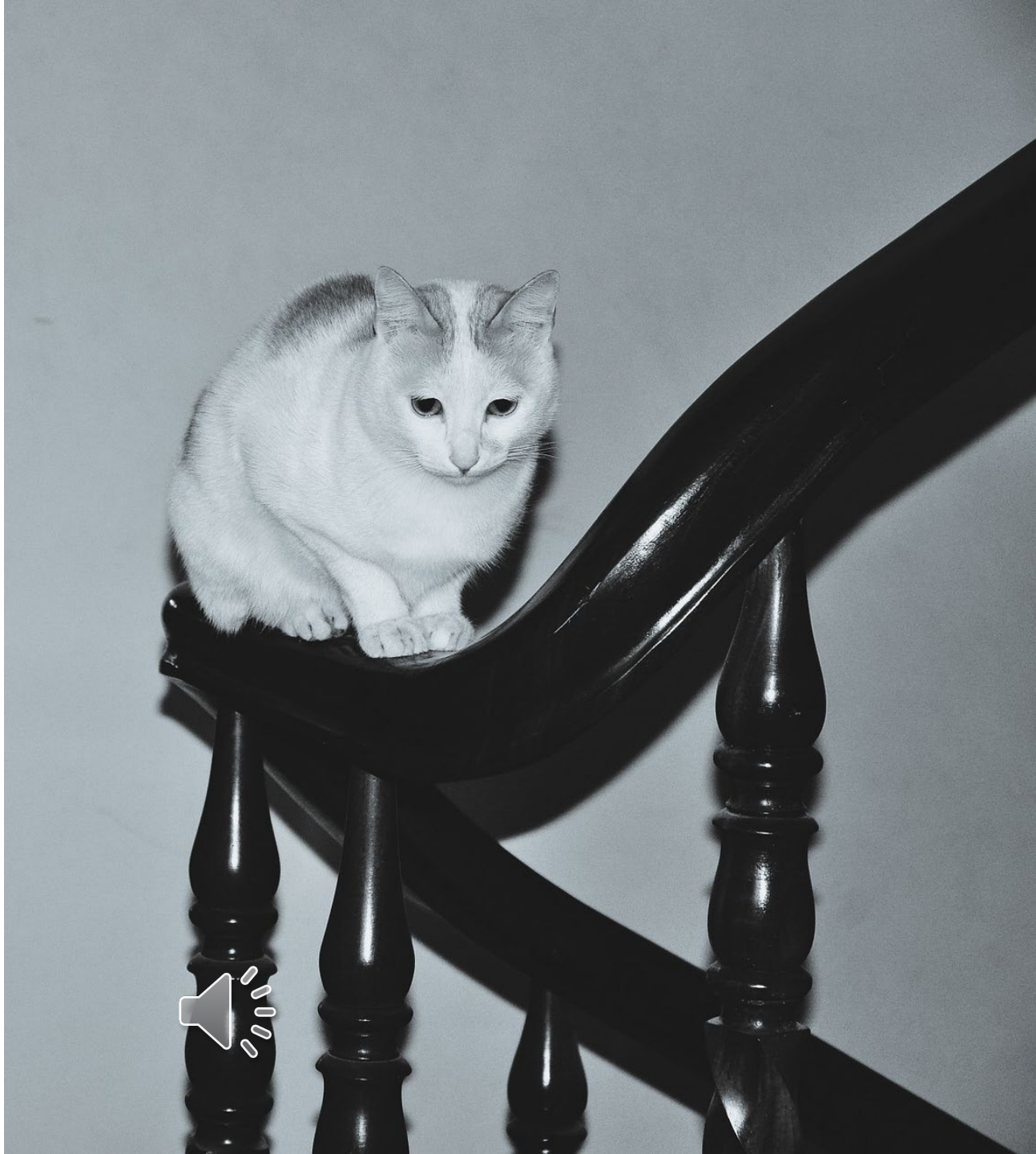
- Define your message
- Decide
  - Which data is relevant
  - What to highlight
  - How to aggregate
  - Precision
- Decide if you need to complement available data

Which one?

- It’s not Bottom-Up ***OR*** Top-Down
- It’s both of them!

…in an iterative way

Till you reach the treasure





Last day we saw the bottom-up approach with the types of data

We will see both bottom-up and top-down approaches with the types of charts



Charts by function

In the top-down approach we start from the audience of the visualization and its objective to select the most appropriate graphic.

In this approach it is important to know the function of each type of chart.

This section is based on the classification of the [Visual Vocabulary, de Financial Times].



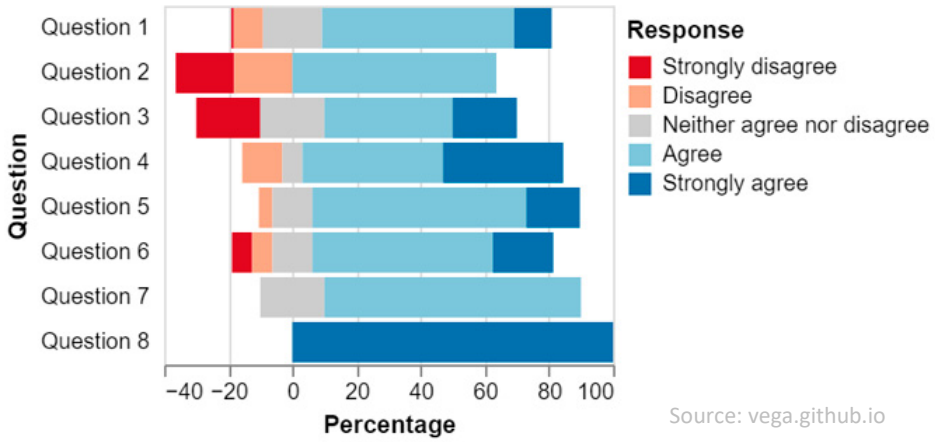
Overview (function)

1. **Deviation**
   Emphasize variations
2. **Correlation**
   Shows the relationship between two or more variables
3. **Ranking**
   Ordinal position is more important than value
4. **Distribution**
   Displays values and their frequency
5. **Evolution over time**
   Emphasize trends
6. **Magnitude**
   Size comparisons
7. **Part of a whole**
   Shows the division of an element into components
8. **Spatial**
   Geographical patterns are the most relevant aspect to display
9. **Flow**
   Intensity between two states

1 Deviation

Stacked divergent bars

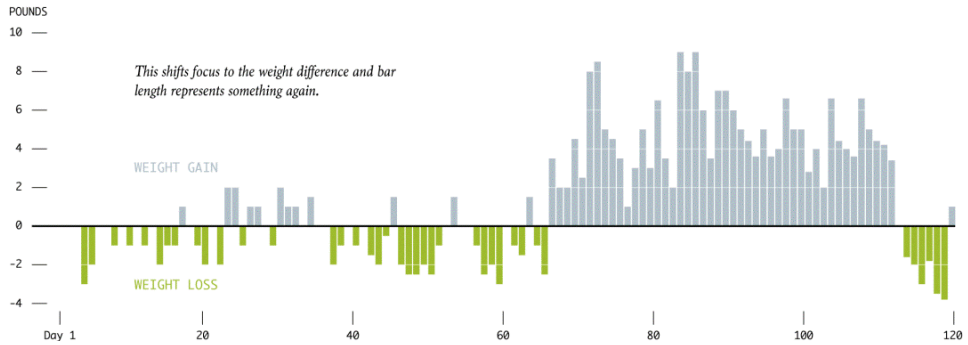
Perfect for opinion polls



Source: vega.github.io

Differential bar chart

A measure separates values into less or greater than it



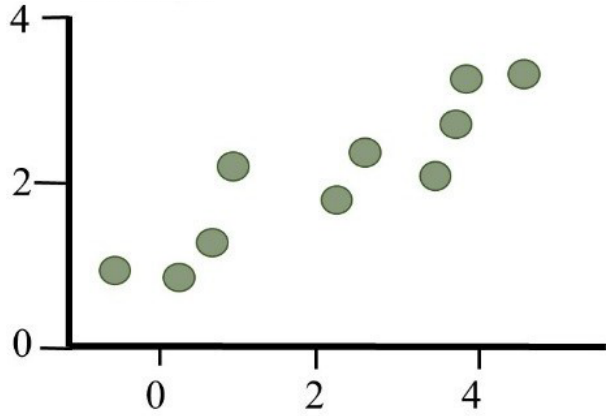
Source: vega.github.io

Emphasizes variations (+/-) from a reference point. The reference point is usually 0 but it can also be another relevant measure. Also used to show opinion (positive/neutral/negative)

2 Correlation

Scatter plot

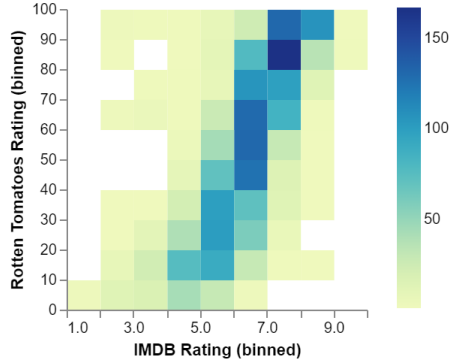
Shows correlation types: positive, negative, nonexistent



Heat map

Ideal for showing patterns.

Enables high data density.



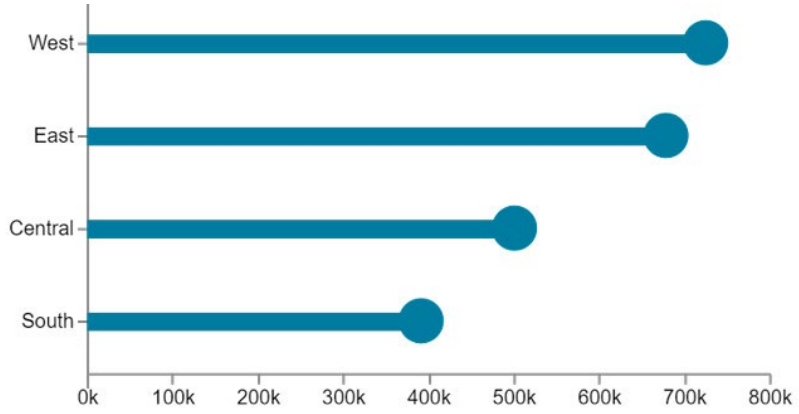
Source: vega.github.io

Correlation plots show the relationship between 2 or more variables. Often readers will assume that it is causal, so it must be made explicit that only correlations are shown.

3 Ranking

Lollipop chart

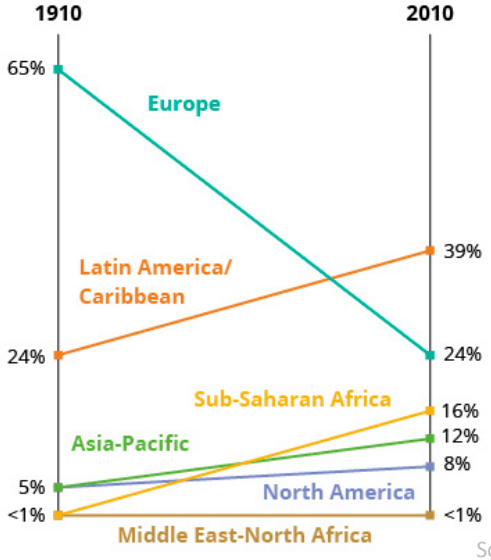
Ordered values with emphasis on the final value



Source: Vega Authors & Community, Gramener

Slope chart

Variation in ranking over time



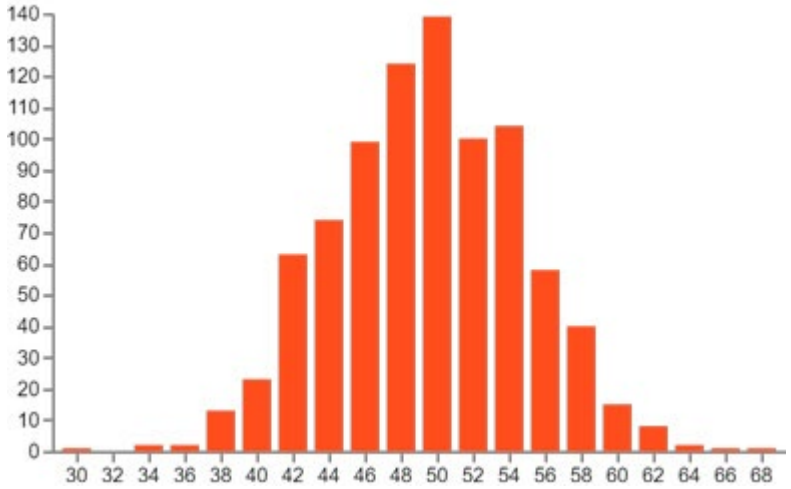
Source: The Huffington Post

They are used if the ordinal position of an item relative to others is more important than its value. Points of interest can be highlighted

4 Distribution

Histogram

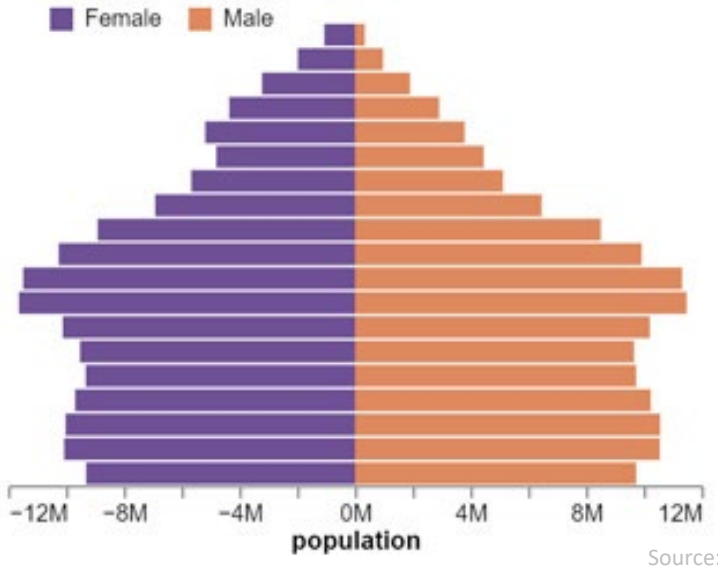
Displays statistical distributions



Source: Vega Authors & Community, Gramener

Butterfly chart

Displays a value according to 2 categories



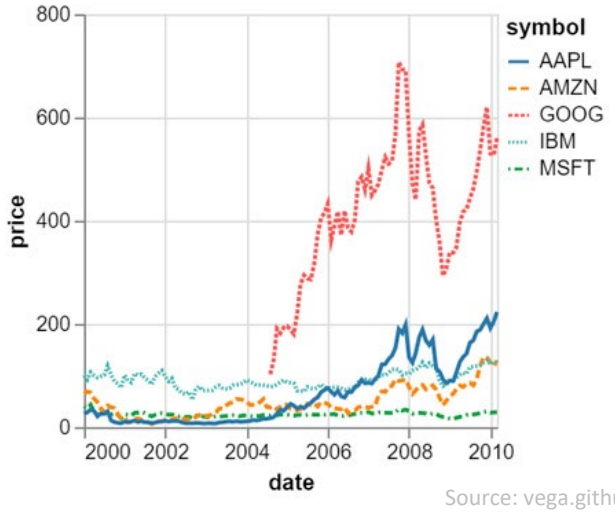
Source: vega.github.io

Displays the values and their frequency. The form (or bias) of the distribution is a way of highlighting the heterogeneity of the data and its normality.

5 Evolution over time

Line chart

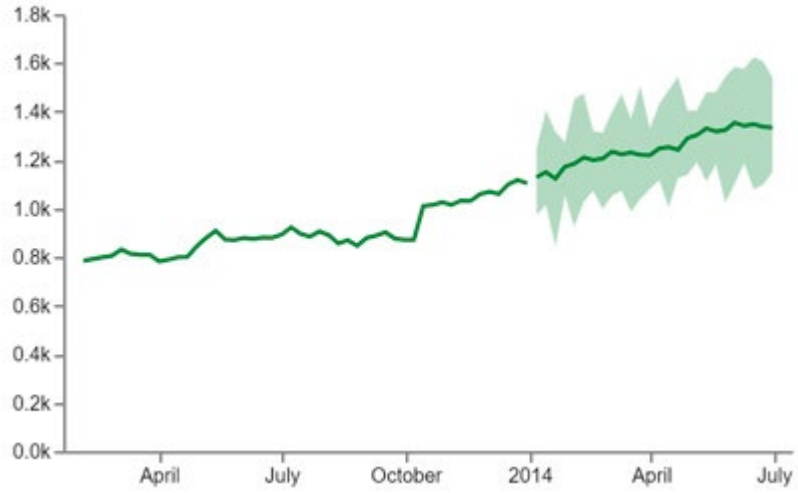
The standard chart for showing changes in a time series



Source: vega.github.io

Fan chart

To show uncertainty



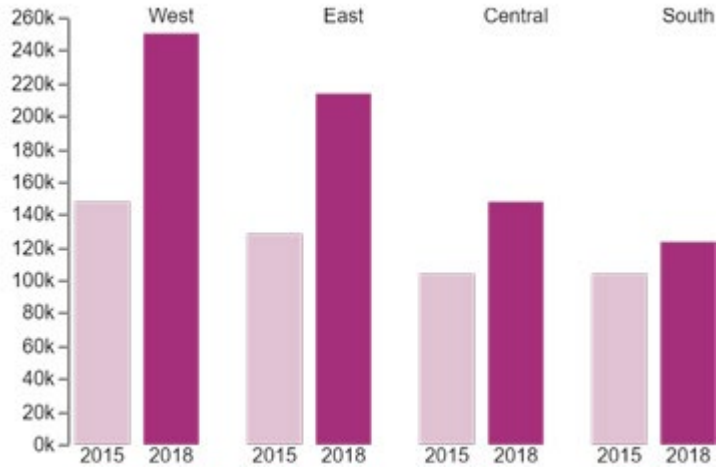
Source: Vega Authors & Community, Gramener

Charts to emphasize trends. Choosing the right period of time is key to providing the reader with the right context.

6 Magnitude

Parallel columns

Perfect for comparison. Maximum 2 categories



Source: Vega Authors & Community, Gramener

Bullet chart

Displays a value in the context of various reference measures



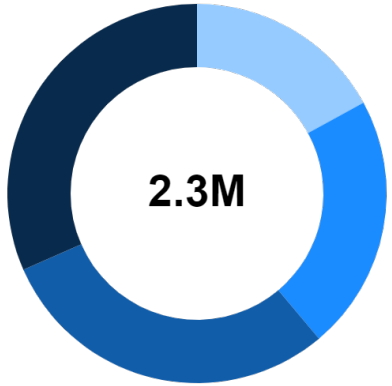
Source: Vega Authors & Community, Gramener

They show relative or absolute size comparisons.
Typically, with discrete values

7 Part of a whole

Doughnut chart

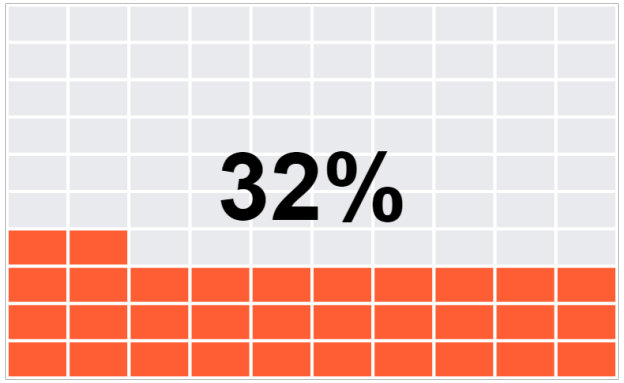
Do not use with many categories. The emphasis is on proportion, not exact measure.



Source: Vega Authors & Community, Gramener

Waffle chart

Ideal for displaying percentages



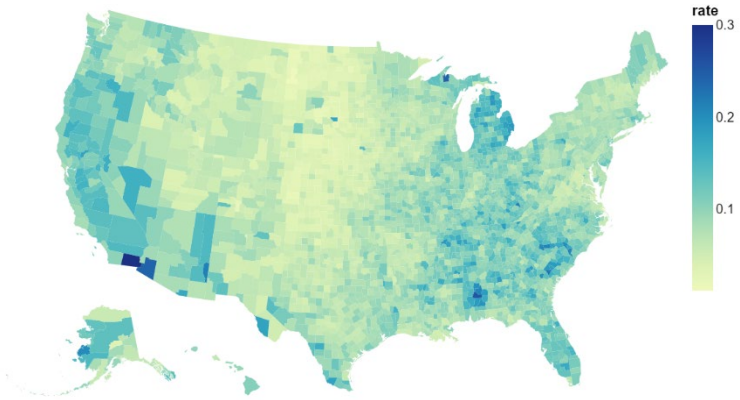
Source: Vega Authors & Community, Gramener

They show how a single element is divided into different components. If the most important thing is the size of the components, better to use a magnitude graph.

8 Spatial

Choropleth map

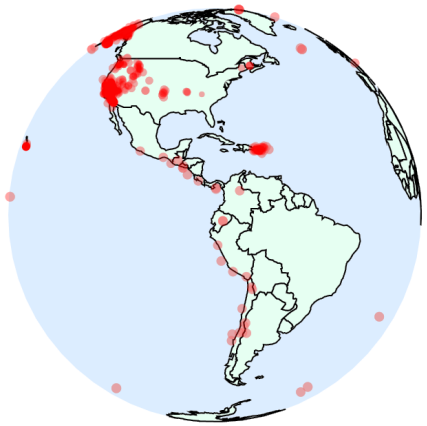
Use with ratio



Source: Vega-Altair

Symbol Map

Absolute values with their exact location



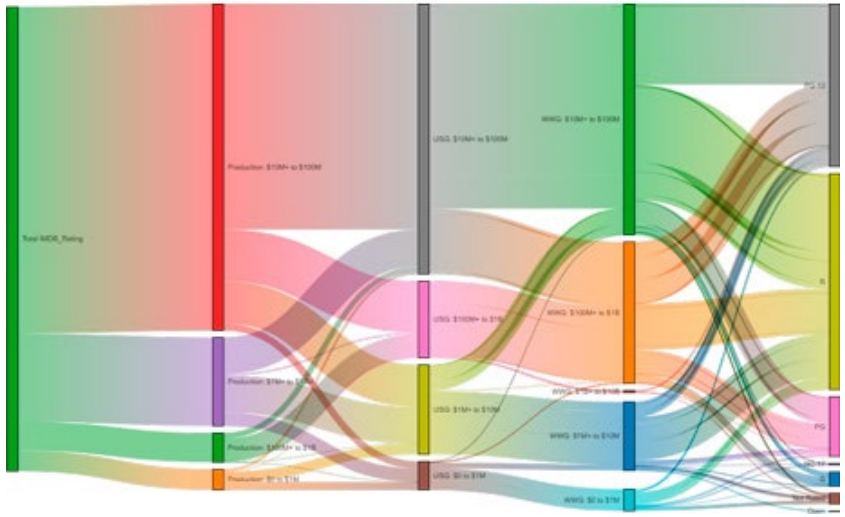
Source: vega.github.io

Use these charts only when there are precise locations or when geographic patterns in the data are most important to the reader.

9 Flow

Sankey diagram

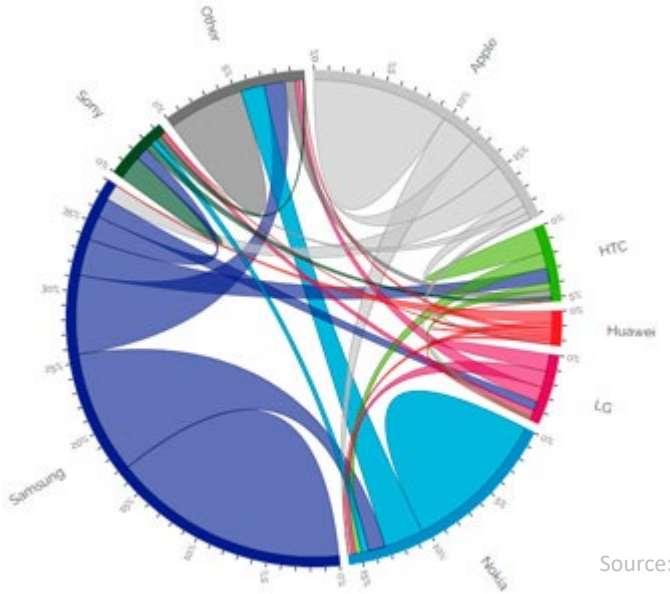
Perfect for process tracking.



Source: Observable, Inc.

Chord diagram

Visually appealing, a bit complex.



Source: Visual Cinnamon

It shows the reader the volumes or intensity between two states or conditions.
These can be logical sequences or geographic locations.

Some well-known graphics

Bar chart

It relies on length to show magnitude. Ideal for comparisons, it should always start at 0.

Cannot display a lot of data at once

1 numeric attribute

1 or more categorical attributes

Line chart

Similar to the scatter plot, it emphasizes the connection of points. The x-axis must be a continuous variable, often time. It's a widely adopted chart.

Scatter plot

It is based on position to show magnitude.

You can add other attributes with size and colors, and be enriched with the regression line. It's a well-known graphic.

1 numeric attribute

1 categorical attribute

If our audience is not very familiar with graphics, we will prioritize highly adopted graphics.

Source: FromDataToViz

Bibliography, types of data

- Frost, J. (2020) Introduction to Statistics: An intuitive guide for analysing data and unlocking discoveries. 1st ed. Pennsylvania: Jim Publishing.

- Brewer C. Color Brewer 2.0

Sources, types of charts

- Financial Times Visual Vocabulary [Internet] https://github.com/Financial-Times/chart-doctor/tree/main/visual-vocabulary

- From Data to Viz [Internet] https://www.data-to-viz.com/about.html

- Visual Vocabulary in Tableau: https://www.tableau.com/solutions/gallery/visual-vocabulary

Thank you and Rewrap





UNIVERSITAT DE BARCELONA

Key ideas when selecting a chart

1. For a visualization to be effective, it will be necessary to know the data, the audience and the data coding techniques in interactive graphic language.

2. The selection of a chart depends on the type of data and its function.

3. Adoption and familiarity is also a criterion to keep in mind.

4. The graphics rely on preattentive properties and Gestalt principles to be very efficient.

How to approach the visualisation:



TOBOGGAN

TOpdown

BOttomUp

and

aGAiN